

Artificial Intelligence: Weapon and Shield in Cybercrime

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August 6th, 2025

Abstract

In the digital age, cybersecurity is a critical concern for individuals, organizations, and governments. With the explosion of data and devices connected to the Internet, traditional cybersecurity measures are increasingly inadequate. Artificial intelligence (AI) promises smarter, faster and more adaptive solutions, such as DarkTrace and Cylance, but it also presents new challenges, such as people exploiting the use of DeepFake technology. This paper investigates how AI is transforming cybersecurity, using real-world cybercrime case studies to explore its advantages and risks.

1 Introduction

Artificial intelligence is a word that defines our technology today. Artificial intelligence, often referred to as AI, is the ability of computers to learn from big data in order to perform tasks and processes that once required human intelligence. With AI playing a dominant role in our daily use of technology, it has provided us with things we couldn't have done in the past decades. In technical fields such as computer science, AI has made data processing much easier and efficient for humans such that AI is ingrained in this field. Artificial intelligence is increasingly being implemented in cyber security, allowing us to detect problems and mitigate attacks in the virtual world.

However, people all over the world have access to AI at their fingertips. With such power, AI can be exploited in ways that cause different types of cybercrime that were never seen before. The efficiency of AI can spread phishing threats at high speeds and to large numbers of people, and the high technology it has can copy images and audio to impersonate identities.

The dual nature of AI in cyber security is an issue discussed in the modern world, especially where technology is an important part of our daily lives. In this research paper, we will discuss how AI plays a role in cybersecurity, both as an offense and as a defense, veering the focus to cybercrimes. We will investigate the role of AI in cybercrime and cyber security, by investigating past cybercrime cases and researching current AI usage for cyber security. With the research findings, we hope to raise awareness about AI-driven cybercrimes and evaluate the effectiveness of AI cybercrime defense.

2 AI and Cybercrime

A fraud occurred Hong Kong in 2024, affecting a company called Arup. Arup is a construction engineering company based in Britain that also focuses on sustainable engineering. A notable building led by this company is the Sydney Opera House. In Arup's Hong Kong office branch, an employee was tricked into sending money to the attackers, who used DeepFake and its AI technology to generate video and audio to impersonate senior officers of the company. This scam caused a loss of around 25 million US dollars to the company.

2.1 DeepFake

This fraud was a social engineering case, where it started as a phishing email, then escalated to using Deepfake technology, impersonating several senior officers of the company. Deepfake technology is video, audio, or visual manipulation that is typically carried out by AI. It relies on AI technology such as autoencoders and generative adversarial networks. Autoencoders aid in compressing and reconstructing input data, allowing efficient processing and understanding of the data. Generative adversarial networks (GANs) is a machine learning framework that puts two neural networks against

each other to generate more realistic fakes. There is face swapping, imitation of expression, and even voices. Deepfakes have been misused by others, not only tricking people to leak information, but also to spread misinformation. Some people use it for harmful purposes like generating harmful videos or content without their consent. Deepfakes can generate realistic videos, making it hard to identify whether it is real or not. They even create images of people who don't exist.

The process of creating deepfakes requires a lot of data collection, collecting a large dataset of videos, images or audio. The more diverse of the data (such as different angles, emotion, movement, lighting conditions, etc.) will create more realistic deepfakes. People can train deepfakes to learn algorithms using neural networks, later on training these collected data. These AI models are inspired by the human brain, effective at pattern recognition and are able to replicate facial features, expressions and voice patterns.

There are two AI models: Autoencoders and Generative Adversarial Networks (GANs). Autoencoders are AI that learns to compress and reconstruct images of a face, used mostly for face-swapping. GANs are two AI systems competing, there is a generator that tries to create fake images, and there is a discriminator that tries to detect if they're fake. Over time, the generator will improve at fooling the discriminator, creating more realistic results. The AI is trained on the collected data and images / video. It learned facial features, voice patterns for audio and how they change with movement. The training can take hours or days depending on the power of the computer and the quality of the data.

During a face swap, the trained model will replace the original face in a video with the target face. For audios AI clones a voice and generates speech from text. In a full synthetic video, the AI model can generate entirely new content, the content can make someone say or do something they never did. Some creators or hackers use video editing software to improve realism, they will edit including matching lighting or shadows, syncing voice and the movement of the lips, add smooth transitions and fixing eye blinks.

2.2 Case Study

In January of 2024, a finance worker first received a phishing email regarding a secret transaction. Although they were suspicious of the email, the person attended a virtual meeting with what seemed like their Chief Financial Officer and other senior officers. However, the meeting consisted of scammers who used Deepfake's AI technology to impersonate not only one, but several members of the meeting. As the employee didn't suspect anything wrong from the meeting, we can infer that the attackers utilized more advanced Deepfake technology. Over these several courses of events, the employee had come to believe this was legitimate and serious matter, preceding to transfer large sums of money to several local bank accounts through many transactions. However, checking with the head office, it was discovered that the event was a scam. This was later reported to the Hong Kong Police. After investigating, the police were able to make 6 arrests of people who were accused of being involved in Deepfake scams. This resulted in an estimate of the loss of 25 million US dollars.

From this case, we learn how AI technology is and has been advancing in the present, but also showcased a different cyber attack. Unlike traditional, major cybercrimes where systems or security were virtually attacked, it utilized a psychological approach. With the help of AI, it allowed one to be manipulated, easily getting past layers of security in the virtual world. In an age where technology is such a huge part of our lives, it is important to also check with the actual person through secure measures, such as meeting up in person or asking other people for verification.

3 AI and Cybersecurity

Although society may perceive AI as a harmful tool, it is still an important aspect in the virtual world. To combat potential cyber attacks and prevent major cyber crimes, AI allows for fast and efficient workflow in the cyber security field. Many softwares are being developed that utilize AI to protect data.

3.1 DarkTrace

DarkTrace is a self-learning AI system that can be used to detect and respond to threats. A self-learning AI system is where AI understands the behavior of the digital environment, such as how

the networks and devices interact and users habits, and applies their understanding to detect any suspicious changes from the users environment. With its multi-layer AI security models, AI detects potential threats primarily through anomaly-based detection. Anomaly-based detection is when a certain habitual baseline is established from the users data, and any deviations from the baseline are flagged as a potential threat. Additionally, DarkTrace finds weak spots and reports to users. AI processes data and the digital environment 24/7, allowing it to report vulnerabilities before being exploited by attackers. If attacked, AI is trained to detect many types of malware attack and neutralize it or alert the security team. In addition the AI keeps user information safe and confidential to prevent ransomware attacks, data breaches, and insider threats. Altogether, DarkTrace is a convenient tool to protect our digital world as AI consistently monitors in search for threats, allowing to thoroughly mitigate threats and attacks in real time.

3.2 Cylance

Cylance, currently called Arctic Wolf Aurora, is another software that uses AI for cyber security purposes. Cylance stands out because of a major cyber attack event called WannaCry, where the AI was trained to protect users from WannaCry and its variants. The WannaCry outbreak was a ransomware attack that affected people worldwide in 2017. However, users who had Cylance installed during the attack were protected from the WannaCry worm. The results prove that Cylance is, in fact, an effective system that utilizes AI and machine learning from early on and manages to keep up to its claims.

Cylance is a self-learning AI system that is designed to detect and prevent cyber threats. Similarly to DarkTrace, Cylance trains on massive datasets to recognize patterns and abnormalities associated with potential malware and threats. Cylance also integrates multiple layers of AI and machine learning models to actively block threats. Not only that, it develops computer software to prevent viruses. What separates Cylance from Darktrace is that unlike Darktrace, the AI in Cylance is actually monitored to make it more secure.

3.3 Simulation

We created a conversational AI on VoiceFlow, a public AI creation website. When talking with this AI, users can choose the cyber dangers and threats they want to ask and identify. If users are completely unfamiliar with a particular area, they can choose to have the AI provide them with relevant knowledge. Initially, without any data, the AI will only provide very short and ambiguous responses because its model and personality were initially unstable. It often strayed from the topic of cybersecurity. Just like educating a child, we provide it with more external exposure, allowing it to gain access to more relevant knowledge and enrich its brain, its database. This database is half sourced from the Internet and half curated by us. We feed the AI with information on prominent cybercrimes we've searched online, as well as cybersecurity measures from government and authoritative organizations, giving it a wealth of background on cybercrime. We also incorporate the cybersecurity knowledge we have learned from teachers and DTCC faculties over the past few weeks, as well as the aforementioned analysis of weaknesses in various cases, into AI to better assist users. Finally, through this approach, our AI is gradually able to provide more constructive advice in conversations with users than before. It uses real-life examples to teach users about cybersecurity, and even the importance of critical thinking, caution, and fraud prevention measures in daily life.

If we had more time, we would like to try to truly utilize this AI. For example, we would build a test website or social software and place our AI there. Every time a user clicks on a phishing link, they will at least have a way to tell them about the threat index and related risks of the website. Despite all the above, the process of building this AI is actually very simple. We don't even have much programming experience. This also shows that AI is gradually becoming more accessible and easy to understand. We also encourage everyone to truly understand AI and explore its potential, whether in cybersecurity or any other field.

4 Future

In today's society, AI and technology are constantly impacting us, everywhere, because AI itself is constantly improving. The AI world is a school, and AI is a learning model, a student just like us. It can reflect on its mistakes and correct them to generate better answers next time. Our role as developers and researchers is to be its teacher, recording and monitoring its achievements and progress. At the same time, there are many different AIs in the world, each is a unique student. Analyzing the strengths and weaknesses of each AI individually is the crucial key for developing more effective AI. Hence, long-term observation and recording of multiple AIs is the best way to understand AI, but unfortunately, we could not yet achieve this. In the future, we hope to invest more time in observing and polishing AI in cybersecurity to establish a more systematic and scientific AI training mechanism. By chronically and multifacetedly documenting the development of different AI models and learning from their strengths and weaknesses, we can empower these once-a-time students to become our future teachers. People shouldn't view AI as a threat that will cost them their jobs, but rather as a valuable tool.

5 Conclusion

With the widespread and easy accessibility of AI in the modern world, it is important to learn the different aspects of what AI can accomplish, both positive and negative. We should fully embrace AI and learn to use it correctly. AI has changed the way we approach cybersecurity as they open tools like DarkTrace and Cylance, which prove how AI can help organizations detect unusual activity, prevent attacks from spreading fast and predict potential threats before it becomes harmful. These systems learn and adapt over time, meaning they can handle new and complex threats faster and more effectively. However, AI is not only used for defense. People have also used it for cybercrimes. One example is Deepfake, where AI creates fake realistic images, videos, or voices that can be used to spread false information, commit fraud or damage reputations. Although AI can greatly improve security, it also introduces new risks that we must prepare for. The future of cybersecurity will rely on using AI wisely, and also continue to develop smart defenses while thinking ahead of the games to be ready for the risk AI can create. We need to understand and consider both benefits and threats, so we can work toward a safer digital world.

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